

One of the most important aspects of an academic career is the education of students. In this pursuit, I seek to (1) provide a conducive and welcoming environment to maximize students' learning and engagement and (2) provide motivated students with opportunities to learn advanced material beyond the classroom and engage in research. In particular, with my background in both theoretical mathematics and applications, I am able to (1) be versatile in the range of courses that I can teach and (2) provide a diverse set of research projects for interested students.

Teaching

Mathematics is not a spectator sport; one cannot simply learn it by watching others do it. Instead, one must engage and wrestle with it themselves. In teaching mathematics, I strive to challenge students to engage directly with mathematics by tackling problems so that they can appreciate and understand the utility and context of what they learn in class. To make this process more approachable and welcoming, I aim to equip my students with the necessary tools to solve problems while also promoting student collaboration. Accordingly, I have experimented with active-learning techniques in my teaching to encourage student engagement and collaboration, and I plan to expand upon them in the future.

Concrete over clean. My first teaching experience was as a teaching assistant (TA) for Math 34A, a calculus course for life-science and social-science majors at UC Santa Barbara. Nearly all students viewed mathematics as something to endure, rather than as something to look forward to.

In one of my first discussion sections, I presented a problem about the position of a car that abruptly accelerates. I went for what I thought was a “natural” approach, which was to construct a piecewise function with separate expressions for before and after the acceleration. This was principled and generalizable, so what could go wrong? Unfortunately, my explanation confused most of the students. My instinct for a clean solution had backfired. The students did not need something clean and tidy; they needed something concrete and accessible. This experience taught me the importance of building intuition and working through examples before introducing formalism and abstraction.

Doing, not watching. As a TA at UCLA and UCSB, I have taught a variety of courses, including calculus (both for majors and for non-majors) and linear algebra (both introductory and proof-based).

I structure my discussion sections around problem-solving. I begin by posing problems for students to work on, encouraging (but never forcing) collaboration so that they can learn from each other's approaches. I circulate to check progress, help students get started, and nudge them in the right direction. In the second half of a discussion section, I discuss solutions, particularly for problems that students find difficult. Rather than jumping straight to an answer, I often pause and ask students to fill in the next step or justify a claim that I had left incomplete.

One of my greatest joys as a TA has been watching students who felt they were struggling gain confidence in their understanding. This has been especially true in calculus courses for non-majors, where many students enter with the belief that mathematics is not for them. Giving students space to solve problems themselves, with guidance when needed, helps transform that belief.

Grounding abstraction. As a Ph.D. student at UCLA, I was the instructor of record for two proof-based upper-division courses. I taught Math 115A (linear algebra) in Winter 2023 and Math 110A (abstract algebra) in Winter 2024. Both courses posed structural challenges that I had to work around.

Math 115A is most UCLA students' first encounter with proofs. However, the syllabus does not explicitly cover proof-writing, as students are expected to learn it on their own. Many students find the course challenging, as it is simultaneously their first exposure to both proofs and abstraction. To help equip my students to succeed, when I taught Math 115A, I spent the first week of the quarter going over set theory and proof-writing. I wanted my students to experience proof-writing on its own before being introduced to new abstract definitions. I also grounded the new abstractions in material that students had seen previously. Students arrive at Math 115A after taking a lower-division linear-algebra course, so I connected abstract notions of vector spaces and linear functions to their concrete matrix counterparts.

Math 110A covers ring theory before students have seen group theory. (Group theory is covered in

Math 110B.) Therefore, I could not rely on the usual definitions — rings as Abelian groups with a second operation and ideals as subgroups with an absorption property — and had to use another foundation to build upon. Fortunately, one was available: the integers. When I taught Math 110A, I began the quarter by introducing properties of the integers (e.g., greatest common divisors (GCD) and modular arithmetic). As the quarter progressed and we discussed more abstract structures, I continually reminded students how these related to the integers and, in particular, how the various proofs were parallel. For example, GCDs in $\mathbb{Z}$ have their analog in principal ideal domains and modular arithmetic becomes quotient-ring arithmetic. I also highlighted that, although intuition from concrete cases helps build understanding, it sometimes fails (e.g., the distinction between prime and irreducible elements in $\mathbb{Z}[\sqrt{-5}]$), which motivates the need for careful abstract definitions.

In both Math 115A and Math 110A, I emphasized the intuition behind key results and used concrete examples to ground abstraction. In office hours and lectures, I frequently asked students why certain results held and about the justification for certain steps. Over time, many students who initially felt uncertain about their abilities became much more confident in their understanding. I saw this transformation most clearly among students who were quiet at the start but later became regular contributors in class. In my teaching evaluations for Math 115A, one student who “was nervous to be taking one of the hardest classes in the math major” expressed their gratitude for “the strong foundation that [I] had helped [them] build for proof-writing.”

Outlook. Between my TA assignments and my two courses as instructor of record, I have taught courses that range from introductory to proof-based upper-division courses. My research bridges pure and applied areas in mathematics, which prepares me to teach across a wide range, including introductory courses such as calculus and linear algebra; proof-based courses such as abstract algebra, real analysis, and topology; applications-oriented courses such as numerical analysis and statistics; and graduate courses in algebra and topology. I also hope to develop new courses related to my research, including topological data analysis (TDA) and networks.

My experience as an educator has reinforced for me the importance of active engagement over passive attendance. I have already seen the benefits of this approach in discussion sections, where hands-on engagement in solving problems helps students solidify key ideas while building confidence in their abilities. In my teaching going forward, I seek to further incorporate active-learning techniques, such as inquiry-based learning, in which students develop key concepts through guided exploration. In this setting, students will regularly work together on problems that build toward key theorems and results, as well as problems that require them to apply those results in unfamiliar contexts. In addition, I plan to explore alternative methods of assessment, especially for courses where project-based work is a natural fit. For example, students can work on projects with assessments in the form of regular check-ins, group evaluations, and individual reflections, culminating in a final report and presentation. This approach gives students feedback throughout the process and engages them in work that builds over a term.

Beyond its inherent benefits for learning outcomes, active learning helps address a new problem. With the proliferation of large language models (LLMs), students can generate solutions to problems with minimal involvement. Therefore, educators must reexamine the role of take-home assignments as tools for engaging students with material and for gauging their understanding. In-class active learning gains new importance; it provides students with opportunities to work through problems together and receive feedback from a human being. To this end, I plan to experiment with a flipped classroom, in which students engage with lectures and readings at home and work through problems in class. Project-based assessments also become more valuable, as regular check-ins and presentations require students to demonstrate understanding in ways that cannot be outsourced easily to an LLM.

Rather than merely challenges to manage, LLMs offer opportunities to enrich learning outcomes. Students can use LLMs to help explain concepts they find difficult or to get feedback on their solutions. Because LLMs are far more available than instructors, they can serve as a helpful supplement, always ready when needed. Rather than banning LLM usage outright, I seek to help students use them in ways

that build their own understanding rather than substitute for it.

Mentoring Students

Mentoring students has been one of the most rewarding aspects of my academic journey. I have been the beneficiary of mentors who have invested in my academic development, and I am motivated to extend that same support to others. My goal in mentoring students is to help them develop skills beyond the classroom while giving them the freedom to explore their own ideas. Whether they continue in academia or move into industry, I want my students to gain the confidence to tackle open-ended problems and the ability to think critically about their own work.

Past experience. As a Ph.D. student, my mentoring grew out of UCLA's Directed Reading Program (DRP), which pairs undergraduates with graduate-student mentors to explore advanced topics in mathematics. Although the program's primary goal is not research, I offered research projects as a possible next step for interested students. I mentored nine DRP students; six of them later transitioned to research projects. I also mentored four students through the UCLA Applied Mathematics REU (with Mason A. Porter). I have continued mentoring students as a postdoc; I currently work with one undergraduate and three master's students.

When mentoring students, I begin by getting to know each student's background and goals, and then I give them an overview of my research to identify projects that fit their interests. From there, I work with them to bridge the gap between their preparation and the demands of a project, helping them find a topic with a feasible learning curve that is also meaningful to them. Projects in TDA, while often requiring reasonably substantial reading and background, offer students a chance to connect abstract mathematics with concrete applications across different domains. TDA often attracts students who enjoy abstract mathematics and are also motivated to see how those ideas can be applied. Projects in opinion dynamics, by contrast, tend to be more accessible to a wider range of backgrounds. The core ideas in opinion dynamics are often easy to grasp, but there is room for projects of varying complexity, both with numerical simulations and theoretical analysis. This flexibility makes opinion dynamics particularly well-suited for mentoring students across different levels of preparation.

One of my most successful mentoring relationships, involving Leila Thompsky and Yuexuan (Yolanda) Wu, began through UCLA's DRP. Leila started reading about opinion dynamics in Fall 2023 and soon wanted to transition to a research project. She brought ideas about what kind of social phenomena to incorporate, and together we brainstormed plausible formulations for an opinion model. Once we settled on a formulation, I guided her through adapting techniques from previous work to develop the theoretical framework of our model. In Spring 2024, my final quarter at UCLA, Yolanda started reading with me and expressed interest in working on similar topics. I connected the two students, and they began collaborating on the project (jointly advised by Mason A. Porter and me).

Leila had made progress on the theoretical side, but the necessary numerical simulations were largely undeveloped before Yolanda joined the project. Leila and Yolanda jointly took up that endeavor. After finishing my Ph.D., I continued meeting with them regularly over Zoom to tighten up theoretical results, review numerical results, and provide feedback on their writing; they are the first two authors on the resulting preprint, which received positive reviews from the journal *Physical Review E*. Both Leila and Yolanda are now pursuing graduate studies. Leila recently started a Ph.D. in applied mathematics at Princeton University, and Yolanda is finishing a master's degree in financial engineering at Carnegie Mellon University. While not every mentoring relationship results in a paper, I find the process with each student to be worthwhile. Students learn to work through open-ended problems and to communicate their findings, which benefit them substantially in both academic and industrial settings.

Outlook. I look forward to continue mentoring students as I advance in my career. Through independent reading, research projects, and general guidance, I hope to provide students with educational opportunities and experiences beyond the classroom. Mentorship is more important now than ever, especially as artificial intelligence (AI) reshapes the academic landscape and, in some cases, seemingly calls

into question the purpose of a mathematical education. Whether in independent readings or research projects, I seek to help students develop genuine understanding and to be deliberate about how they use AI. In an independent reading, this may involve using AI to clarify a concept, rather than skipping through important derivations. In a research project, this may involve using AI to help create and refine figures while keeping the conceptual work their own. As students adapt to shifting paradigms, I plan to offer them what I have learned and the reassurance that they are not working through it alone, even as I adapt to these changes myself.

As my research expands, I seek to work with students from a wide range of interests and backgrounds, including those outside of mathematics. Two of the four students that I currently mentor come from computational social science, rather than mathematics. Working with them has broadened my perspective on interdisciplinary research. As a faculty member, I plan to run a regular seminar where students can present their work and discuss recent literature with their peers and other researchers. Mentoring students is a priority that I will carry with me throughout my career, and I am eager to help students find their own paths in research.